# A REDUCED LOCALLY NOETHERIAN SINGLETON ALGEBRAIC SPACE IS REGULAR

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*Editorial note.* Counted span is the author's own proof environment, source lines 3338-3351, immediately following the statement at 3333-3336.

*Editorial note.* One cross-chapter pointer (Descent) and two same-chapter line-indexed stubs (lemma-unique-point-better, remark-recall) are recorded in assembly.external_references.

*Allowed standard prerequisites (editorial).*

- Existence of a flat fppf point on a singleton space. Lemma lemma-unique-point-better of this chapter, cited at proof lines 3341-3343. Supplies the surjective flat locally finitely presented morphism Spec(k) -> Z from a field that the fibred product needs.
- Regular schemes descend along fppf morphisms. Descent, Lemma descent-lemma-descend-regular, cited at proof lines 3347-3348. T -> W flat, surjective, locally of finite presentation with T regular gives W regular.
- Etale over a field is regular. Remark remark-recall of this chapter, cited at proof lines 3344-3345. T = W x_Z Spec(k) is etale over k, hence regular.

*Source reference.* Original reference 06QN: focused-seeds/geometry/S9-decent-residual-space-regular/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 06QX: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 3135–3135. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original remark 03II: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 459–459. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *A reduced, locally Noetherian singleton algebraic space $Z$ is regular.*

*Proof.* Let $Z$ be a reduced, locally Noetherian singleton algebraic space over a scheme $S$. Let $W \to Z$ be a surjective étale morphism where $W$ is a scheme. Let $k$ be a field and let $\mathrm{Spec}(k) \to Z$ be surjective, flat, and locally of finite presentation

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(see Lemma 06QX). The scheme $T = W \times_Z \mathrm{Spec}(k)$ is étale over $k$ in particular regular, see Remark 03II. Since $T \to W$ is locally of finite presentation, flat, and surjective it follows that $W$ is regular, see Descent, Lemma 06QN. By definition this means that $Z$ is regular. $\square$